وثيقة المراجعة والتلخيص الأكاديمي

ASE · Chapter 5 · Software Design — Aggarwal & Singh, 3rd ed. (Bloom Level 6)

Source: slide deck Ch05_Software_Design.pdf (99 slides) + printed book pp. 202–251. Case study = **Library Management System** (slides 81–93; book pp. 236–247). Chapter data tables: slide 19 (coupling), slide 27 (cohesion), slide 57 (SDD Table 2), slide 72/75 (OOD steps, 6 boxes), slide 75 (object types), slides 77–78 (class relationships), slides 84–93 (case-study artefacts).

Format: English question + full worked artefact in English; one short Arabic justification per item. Every sub-question states its assumptions, produces the artefact (tables / indented trees / ASCII-structured lists), then justifies. Blank lines separate every block (RTL-safe). No frontmatter.

Two frozen facts (must not be contradicted anywhere in this file):

Coupling best → worst = **Data** > **Stamp** > **Control** > **External** > **Common** > **Content** (Fig. 7, slide 19 / book p. 208–209).

Cohesion best → worst = **Functional** > **Sequential** > **Communicational** > **Procedural** > **Temporal** > **Logical** > **Coincidental** (Fig. 11, slide 27 / book p. 211).

OOD = exactly **SIX** steps in this order (slide 72 Fig. 25 / book p. 232): (1) Create use case model → (2) Draw activity diagram (if required) → (3) Draw the interaction diagrams → (4) Draw the class diagram → (5) Draw the state chart / object diagram (if required) → (6) Draw component & deployment diagram → design documents.